DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2021/2022

THIRD YEAR **SEMESTER** 2 EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS INFORMATION TECHNOLOGY
CIT3210: RESEARCH METHODOLOGY

DATE: **TIME:**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

SAMTEC is an upcoming software developing company that has been into operation for the past three years. Tom has been contracted by this company for the position of a research manager. One of the leading telecommunications companies in the country has been going through business loss because they don't have a software to monitor their customers call rates. This company approaches SAMTEC to develop an analysis tool to monitor call rates for their customers. As a research manager Tom is required to develop this software and prepare documentation to your company as a guide on this project.

a) Explain Toms research motivation type (4 Marks)
b) What research type is this? Justify your answer (4 Marks)

c) Discuss one measurement scale which Tom must use in analyzing his variable (2 Marks)
d) Explain one method of data collection which will be appropriate for this research. (4 Marks)
e) What will be his appropriate data sampling technique? Justify your answer. (4 Marks)
f) Explain any four ethical considerations Tom should consider before starting his work (8 Marks)
g) In data analysis differentiate negatively skewed from positively skewed distribution (4 Marks)

Question 2: (20 Marks)

The data below was collected to investigate the spread of a certain virus infection among different fifty five animal species in a town. The data shows age of the animal infected by the disease.

40	29	27	18	44
29	41	34	45	32
21	17	56	22	55
55	32	61	26	67
43	28	13	50	34
13	45	15	33	43
29	23	18	54	32
21	17	23	14	23
56	48	32	34	21
45	55	51	23	22
67	21	55	16	42

a) Calculate the Mean, Mode, Median, Standard deviation and range (10 Marks)
b) Draw a histogram to represent the above data where the x axis should represent class boundaries and the y axis to represent the frequencies of distribution (6 Marks)
c) Use the Histogram in part b. to explain the type of data distribution (4 Marks)

Question 3: (20 Marks)

a) You have just graduated and in a group of seven have decided to open company by the name CUSTTEC research institute. A client who is a donor at a certain school would like you to do research to identify if the school has been misusing the funds donated to support the students.

i. Explain three research tensions related with your work (6 Marks)
ii. In relation to your research, use examples to distinguish a variable from a concept (4 Marks)
iii. In this research, given the concepts needy and best student, come up with at least two clear indicators, two variables and two working definitions that will be used to measure these concepts. (6 Marks)
iv. Explain logical and empirical as characteristic of a good research work (4 Marks)

Question 4: (20 Marks)

a) Distinguish between Conceptual vs. Empirical research types (4 Marks)
You are a research analyst and over the past five years you have been receiving awards as a best performer from your employer. You have been given the task of training new employees that have just been recruited by your company on best practices of conducting research. One of the learners would like to know the difference between nominal and interval scales of data rating, Explain. (4 Marks)
b) In your training sessions explain to the learners the below research methods(10 Marks)

i. Longitudinal Studies
ii. Cross-sectional Studies
iii. Ethnography
iv. Participative Enquiry
v. Grounded Theory

c) State the difference between open and closed questionnaires (2 Marks)